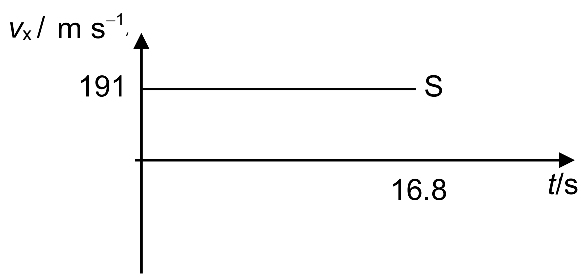
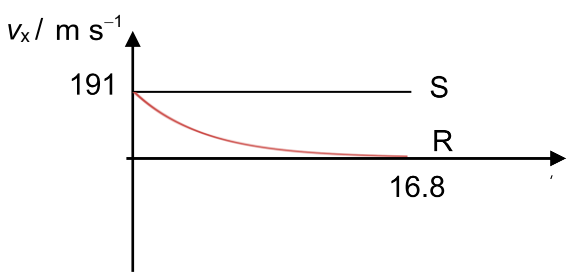
| Qn | Suggested Solutions |
|---|---|
| **2(a)(i)** | $s_y = u_y t + \frac{1}{2} a_y t^2$ <br> $x = 220\sin(30^\circ)(16.8) + \frac{1}{2}(9.81)(16.8)^2$ |
| | $x = 3230$ m |
| **(ii)** | $v_x = u_x = 220\cos(30^\circ) = 190.5 \text{ m s}^{-1}$ |
| | $v_y = u_y + a_y t$ <br> $v_y = 220\sin(30^\circ) + 9.81(16.8) = 274.8 \text{ m s}^{-1}$ |
| | $v = \sqrt{{v_x}^2 + {v_y}^2}$ <br> $v = \sqrt{190.5^2 + 274.8^2}$ <br> $v = 334 \text{ m s}^{-1}$ |
| | OR |
| | initial $E_k = \frac{1}{2} m (220)^2$ |
| | loss in gravitational $E_P = mgh$ <br> $= m(9.81)(3232)$ |
| | final kinetic energy = loss in gravitational potential energy + initial kinetic energy <br> $\frac{1}{2} m v^2 = m(9.81)(3232) + \frac{1}{2} m (220)^2$ <br> $v = 334 \text{ m s}^{-1}$ |
| **(iii)** |   |
| **(b)** |   |